



VAPO

Value-Model-Based Reinforcement Learning
for Advanced Reasoning



60.4 on AIME 2024_{avg@32}

Surpassing DAPO (50) and DeepSeek-R1-Zero (47) by 10+ points

60

Why Value Models? Finer Credit Assignment for Long CoT Reasoning

Value models enable token-level credit assignment — critical when subtle step-level errors cause catastrophic failures in long reasoning chains.

Value-Model-Free (GRPO, DAPO)

Advantage Computation

Based on final trajectory reward
Assigned uniformly across all tokens

Limitations for Complex Reasoning

- ✗ No per-token credit — cannot trace which step caused failure
- ✗ Higher variance from Monte Carlo estimates

- ✗ Limited generalization across samples

Token-level advantage:

Uniform advantage across all tokens



Value-Model-Based (VAPO)

Advantage Computation

Token-level value estimation via learned $V(s)$
GAE computes per-token advantages

Advantages for Complex Reasoning



Precise token-level credit assignment



Lower-variance estimates per token



Generalizes across samples → better sample efficiency

Token-level advantage:



Differentiated advantage per token (error step highlighted)

Three Challenges Plaguing Value-Model-Based RL in Long-CoT Tasks

These three challenges explain why value-model-based methods have underperformed value-free approaches — until VAPO.

1 Value Model Bias

Root Cause

Long trajectories amplify bootstrapping errors in value estimation, leading to compounding bias across the entire reasoning chain.

Impact

Vanilla PPO collapses to a score of just **5 on AIME 2024**

$V(s_1) \rightarrow V(s_2) \rightarrow \dots \rightarrow V(s_n)$

Error compounds with chain length

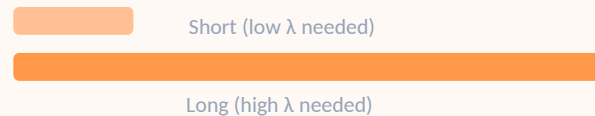
2 Heterogeneous Lengths

Root Cause

Response lengths vary widely (100s to 10,000s of tokens). Short and long sequences need different bias-variance tradeoffs in GAE.

Impact

Fixed λ in GAE is suboptimal; one size does not fit all.



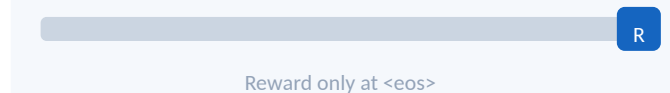
3 Reward Signal Sparsity

Root Cause

Only the final <eos> token receives the reward signal. Thousands of intermediate tokens get no direct supervision.

Impact

Exacerbated by long CoT; harder to learn from failures.



VAPO Framework: Integrated Solutions for Value-Model-Based Reasoning RL

Each technique in VAPO is mapped to a specific challenge — no component is ad hoc.

Challenge 1: Value Model Bias



Value-Pretraining

Pre-trains the value model before RL to reduce initial estimation bias.

Impact: +49 points



Decoupled-GAE

Separates GAE computation for more robust value estimation.

from VC-PPO

Challenge 2: Heterogeneous Lengths



Length-Adaptive GAE

NOVEL

Dynamically adjusts λ parameter based on response length:

- Short responses $\rightarrow$ lower λ ($\downarrow$ variance)
- Long responses $\rightarrow$ higher λ ($\downarrow$ bias)

Optimizes bias-variance tradeoff across heterogeneous lengths.

Impact: +15 points

Challenge 3: Reward Sparsity

Clip-Higher

Asymmetric clipping encouraging exploration
from DAPO

Token-level Loss

Normalizes loss at token level, not sequence
from DAPO

Positive Example LM Loss

Self-imitation learning on correct trajectories

Group-Sampling

Multiple responses per prompt for baseline est.
from GRPO

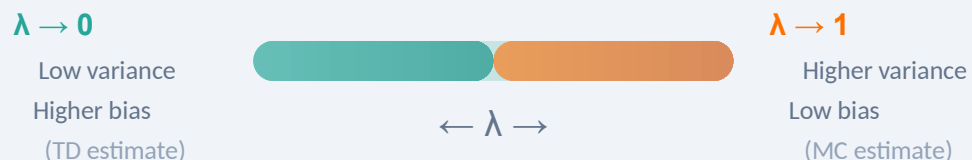
VAPO = PPO + Value-Pretraining + Decoupled-GAE + Length-Adaptive GAE
+ Clip-Higher + Token-level Loss + Positive Example LM Loss + Group-Sampling

Length-Adaptive GAE: Dynamic λ Adjustment for Variable Response Lengths

Novel contribution: dynamically adapts the GAE λ parameter to response length, contributing 15 points in ablation.

The Core Insight

In GAE, the λ parameter controls the bias-variance tradeoff:



Short Responses



Lower lambda

- Few tokens $\rightarrow$ value model estimates are accurate
- Prioritize low variance
- Trust bootstrapped values

Long Responses



Higher lambda

- Many tokens $\rightarrow$ bias compounds over chain
- Prioritize low bias
- Rely more on MC returns

Why This Matters

Long-CoT tasks produce responses ranging from hundreds to tens of thousands of tokens.

A fixed λ forces a single tradeoff point that is suboptimal for both extremes.

Ablation Impact

Removing Length-Adaptive GAE drops AIME score from **60 $\rightarrow$ 45 (-15 points)**

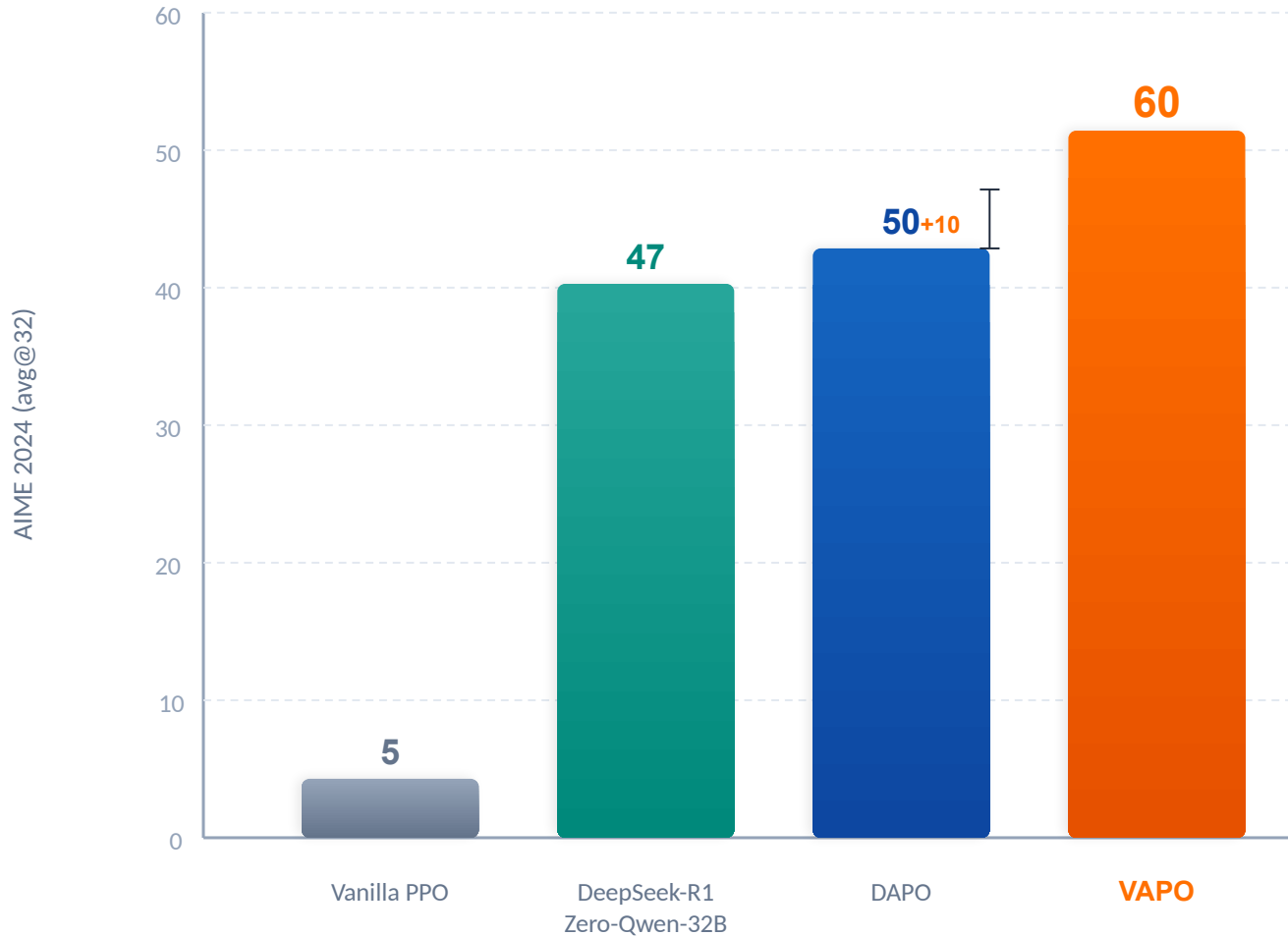
This is the third most impactful component after Value-Pretraining and Decoupled-GAE.



Key novelty: Length-Adaptive GAE is VAPO's original contribution — the first technique to address heterogeneous sequence lengths in value-model-based RL for long chain-of-thought reasoning.

VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO and DeepSeek-R1-Zero

All models use the same Qwen2.5-32B base with no SFT data. VAPO achieves 60.4 avg@32, a new state-of-the-art.



Key Results

60.4 AIME 2024 avg@32
New state-of-the-art on Qwen2.5-32B

- Surpasses DAPO by **+10 points**
- Surpasses DeepSeek-R1-Zero by **+13**
- Vanilla PPO scores only 5 (12x worse)

Experimental Setup

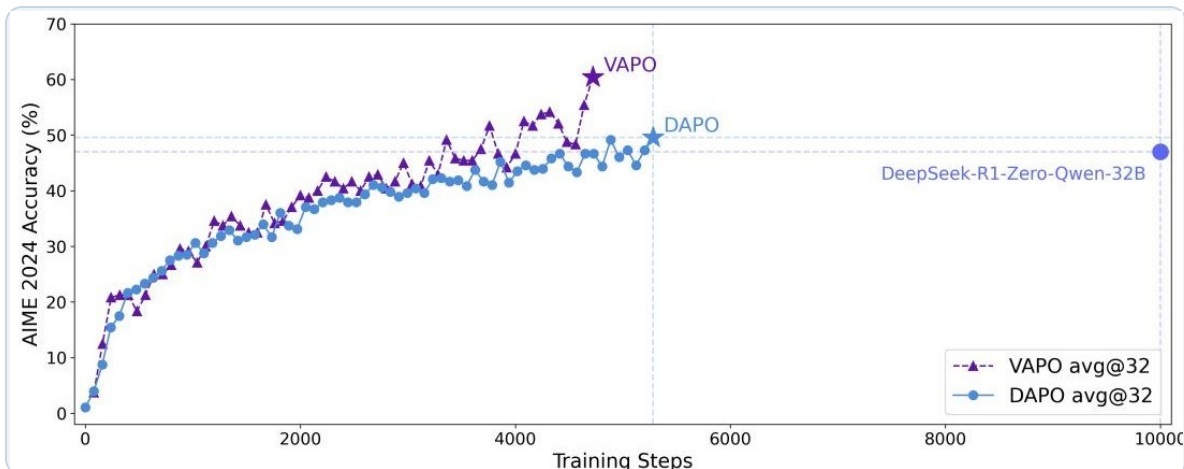
- Base model: Qwen2.5-32B
- No supervised fine-tuning data
- Pure RL training from scratch
- Evaluation: avg@32 (32 samples)

See next slide for training curves showing VAPO consistently outperforming DAPO throughout training.

VAPO Reaches SOTA in 5K Steps — Superior Training Efficiency and Stability

VAPO reaches 60% accuracy at ~4,500 steps with zero training crashes across multiple independent runs.

AIME 2024 Accuracy Over Training Steps



- VAPO reaches **60% at step ~4,500**
- DAPO plateaus near **50% at step ~5,500**
- DeepSeek-R1-Zero reaches 47% after ~10,000 steps

Mean Response Length Over Training



- VAPO generates longer responses:
~ **6,000 tokens** vs DAPO's ~4,500
- Suggests more elaborate reasoning chains



Training Steps to SOTA

~5,000 vs DAPO ~5,500



Training Crashes

Zero across multiple runs



Peak Response Length

~6K tokens (vs ~4.5K DAPO)

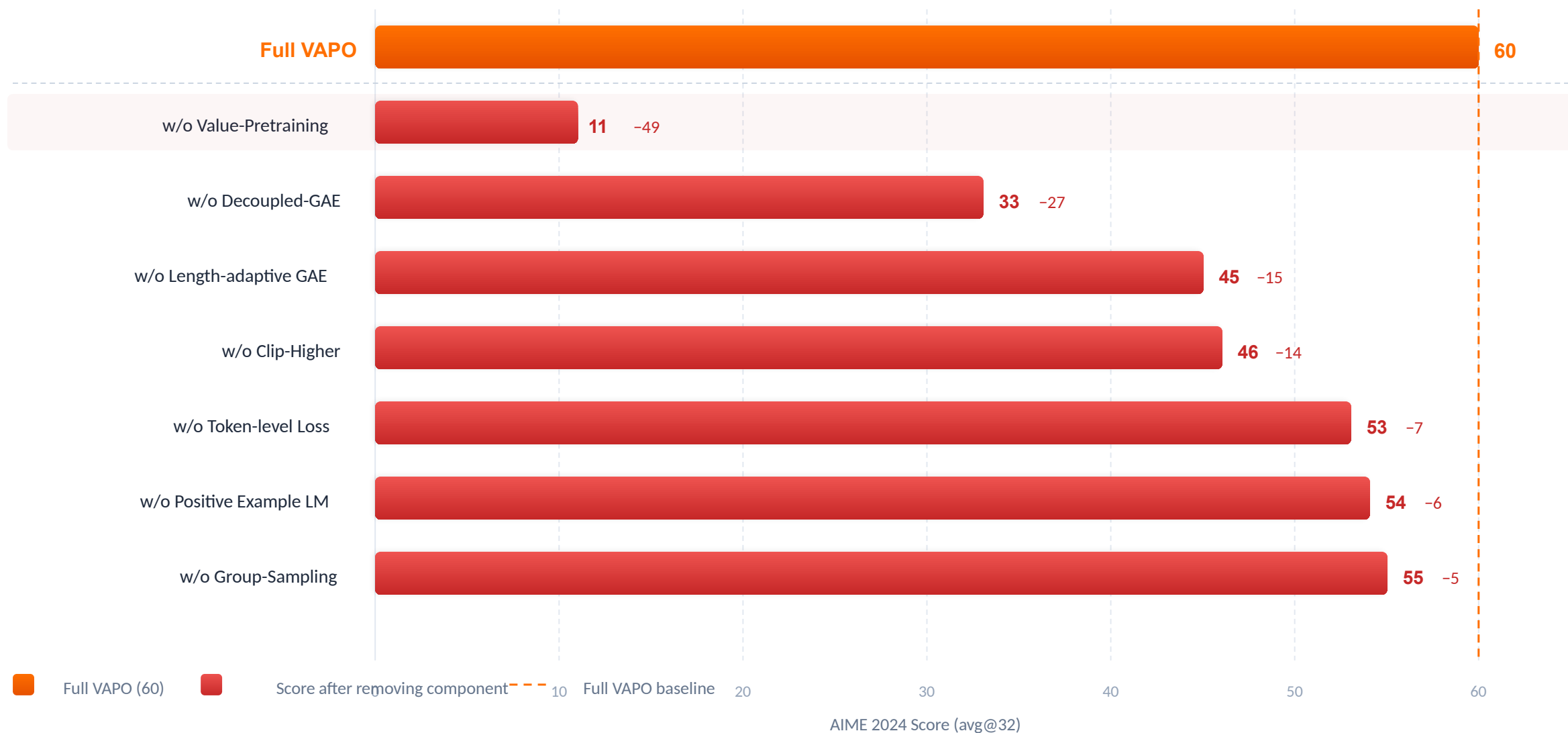


Score Gain vs DAPO

+10 points

Ablation: Every Component Matters — Value-Pretraining Alone Adds 49 Points

Removing Value-Pretraining catastrophically drops AIME score from 60 to 11 — the value model is useless without proper initialization.



Key Takeaways

Value-Model-Based RL Is Back — With the Right Engineering

60.4

on AIME 2024 — surpassing DAPO (50) and DeepSeek-R1-Zero (47)

New SOTA on Qwen2.5-32B base model with no SFT data, pure RL training.

+49

points from Value-Pretraining alone — the single most critical component

Without it, the score drops catastrophically from 60 to 11. Value models are useless without proper initialization.

+15

points from Length-Adaptive GAE — a novel technique for heterogeneous lengths

Dynamically adapts λ to response length, addressing a problem unique to long chain-of-thought reasoning.



Zero training crashes across multiple independent runs

VAPO reaches SOTA in ~5,000 steps — highly stable and training-efficient.



Value-model-based RL can outperform value-free methods for complex reasoning

When the three core challenges (bias, heterogeneous lengths, reward sparsity) are systematically addressed.